The Chain Rule

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Basic chain rule applications**

Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$

$= 5(4x - 1)^4(4) = 20(4x - 1)^4$

b $f(x) = \sin(3x^2)$ $= 6x\cos(3x^2)$

c $f(x) = e^{5x}$ $= 5e^{5x}$

d $f(x) = \ln(2x + 7)$ $= \frac{2}{2x + 7}$

(a) $= 5(4x - 1)^4(4) = 20(4x - 1)^4$; (b) $= 6x\cos(3x^2)$; (c) $= 5e^{5x}$; (d) $= \frac{2}{2x + 7}$

2 **[Foundational]** • **Target: Basic chain rule applications**

Differentiate $f(x) = \sqrt{x^2 + 9}$.

$f'(x) = \frac{2x}{2\sqrt{x^2 + 9}} = \frac{x}{\sqrt{x^2 + 9}}$.

3 **[Foundational]** • **Target: The chain rule with trig and exponential functions**

Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).

$f'(x) = 3\cos^2 x \cdot (-\sin x) = -3\cos^2 x \sin x$.

4 **[Foundational]** • **Target: Nested (multiple) chain rule**

Differentiate $f(x) = \sin\left(\sqrt{x^2 + 1}\right)$, applying the chain rule twice.

$f'(x) = \cos\left(\sqrt{x^2 + 1}\right) \cdot \frac{x}{\sqrt{x^2 + 1}}$.

5 **[Foundational]** • **Target: Chain rule with tables of values**

Given $f(2) = 5$, $f'(2) = 3$, $g(2) = 7$, $g'(2) = -1$, find $\frac{d}{dx}[f(g(x))]$ at $x = 2$ if $g(2) = 7$ and $f'(7) = 4$ (note the composition uses f' evaluated at $g(2) = 7$, not at 2).

$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$; at $x = 2$: $f'(7) \cdot g'(2) = 4 \cdot (-1) = -4$.

6 **[Foundational]** • **Target: Finding tangent lines using the chain rule**

Find the equation of the tangent line to $f(x) = (2x - 1)^3$ at $x = 1$.

$f(1) = 1^3 = 1$. $f'(x) = 3(2x - 1)^2(2) = 6(2x - 1)^2$; $f'(1) = 6(1)^2 = 6$. Tangent: $y - 1 = 6(x - 1)$, i.e. $y = 6x - 5$.

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7 [Foundational] • Target: Visualizing a composite function and its derivative

The graph shows $f(x) = (x-1)^2$ (an inner-outer composite of $u = x - 1$ and u^2), with its tangent line at $x = 2$.

a Use the chain rule to find $f'(x)$. $f'(x) = 2(x-1)(1) = 2(x-1)$.

b Confirm $f'(2)$ matches the slope of the tangent line shown. $f'(2) = 2(1) = 2$, matching the tangent line $y = 2x - 2$, which has slope 2.

(a) $f'(x) = 2(x-1)(1) = 2(x-1)$.; (b) $f'(2) = 2(1) = 2$, matching the tangent line $y = 2x - 2$, which has slope 2.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The chain rule in applied rate problems

The volume of a sphere is $V = \frac{4}{3}\pi r^3$, and the radius is growing according to $r(t) = 2t + 1$ (cm, seconds). Use the chain rule to find $\frac{dV}{dt}$ at $t = 2$.

$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt} = 4\pi r^2 \cdot 2$. At $t = 2$: $r = 5$, so $\frac{dV}{dt} = 4\pi(25)(2) = 200\pi \approx 628.3$ cm^3/s.

9 [Intermediate] • Target: Chain rule with a table of composed values

Given $h(x) = f(g(x))$ where $g(1) = 3$, $g'(1) = 2$, $f(3) = 7$, $f'(3) = -4$, find $h'(1)$.

$h'(x) = f'(g(x)) \cdot g'(x)$: $h'(1) = f'(g(1)) \cdot g'(1) = f'(3) \cdot 2 = (-4)(2) = -8$.

10 [Intermediate] • Target: Chain rule with a table of composed values

Differentiate $f(x) = \sqrt[3]{x^2 + 1}$ using the chain rule.

$f(x) = (x^2+1)^{1/3}$: $f'(x) = \frac{1}{3}(x^2+1)^{-2/3}(2x) = \frac{2x}{3(x^2+1)^{2/3}}$.

11 [Intermediate] • Target: Basic chain rule applications

[Unit 8 · Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

12 [Intermediate] • Target: The chain rule with trig and exponential functions

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

Express $y = (x^4 + 3x^2 + 7)^{1/2}$. By the Chain Rule: $y' = \frac{1}{2}(x^4 + 3x^2 + 7)^{-1/2} \cdot \frac{d}{dx}(x^4 + 3x^2 + 7) = \frac{4x^3 + 6x}{2\sqrt{x^4 + 3x^2 + 7}} = \frac{2x^3 + 3x}{\sqrt{x^4 + 3x^2 + 7}}$.
Evaluate at $x = 1$: $y'(1) = \frac{2(1)^3 + 3(1)}{\sqrt{1^4 + 3(1)^2 + 7}} = \frac{5}{\sqrt{11}} = \mathbf{\frac{5\sqrt{11}}{11}}$.

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13 **[Intermediate]** • **Target: Nested (multiple) chain rule**

[Unit 6 · Integration and Accumulation of Change · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

14 **[Intermediate]** • **Target: Chain rule combined with product/quotient rules**

[Unit 7 · Differential Equations · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Chain rule with tables of values** • **Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

Step 1 · Initial Verification: As $x \to 0$, numerator $0 - 0 = 0$ and denominator $0^3 = 0$. Form $\frac{0}{0}$. Differentiating numerator and denominator: $\lim_{x \to 0} \frac{1 - \frac{1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{\frac{(1+x^2)-1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{x^2}{3x^2(1+x^2)}$. Step 2 · Algebraic Simplification: For $x \neq 0$, cancel the common factor x^2: $= \lim_{x \to 0} \frac{1}{3(1+x^2)} = \frac{1}{3(1+0)} = \mathbf{\frac{1}{3}}$.

16 **[Challenging]** • **Target: Chain rule with inverse trig and logarithms** • **Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

Level 3 · Advanced / AP FRQLet $f(x)$ be a function whose third derivative satisfies $|f'''(x)| \leq 6$ for all $x \in [0, 0.4]$. If the second-degree Taylor polynomial $P_2(x)$ centered at $c = 0$ is used to approximate $f(0.2)$, find the maximum possible error in this approximation using the Lagrange Error Bound.

Using the Lagrange Error Bound formula with $n = 2$, $c = 0$, $x = 0.2$, and $M = 6$: $|R_2(0.2)| \leq \frac{M}{(2+1)!}|x - c|^{2+1} = \frac{6}{3!}|0.2 - 0|^3$. Since $3! = 6$: $|R_2(0.2)| \leq \frac{6}{6}(0.2)^3 = 1 \times 0.008 = \mathbf{0.008}$ $\left(\text{or } \frac{1}{125}\right)$. The maximum truncation error is guaranteed to be no greater than 0.008.

17 **[Challenging]** • **Target: Finding tangent lines using the chain rule** • **Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Level 1 · FoundationsFor the parametric curve $x(t) = \cos(t)$ and $y(t) = \sin(2t)$, find all values of $t \in [0, \pi]$ where the curve has a vertical tangent line.

A vertical tangent occurs when $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0$. Compute $x'(t) = -\sin(t) = 0 \Rightarrow t = 0, \pi$. Check $y'(t) = 2\cos(2t)$: • At $t = 0$: $y'(0) = 2\cos(0) = 2 \neq 0$. • At $t = \pi$: $y'(\pi) = 2\cos(2\pi) = 2 \neq 0$. Therefore, vertical tangents occur at $\mathbf{t = 0}$ and $\mathbf{t = \pi}$.

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18 [Challenging] • Target: Visualizing a composite function and its derivative • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x\to 0}\frac{\sqrt{1+\tan x}-\sqrt{1+\sin x}}{x^3}$.

Step 1 · Rationalize the Numerator: Multiply by $\sqrt{1+\tan x}+\sqrt{1+\sin x}$:

$\lim_{x\to 0}\frac{(1+\tan x)-(1+\sin x)}{x^3(\sqrt{1+\tan x}+\sqrt{1+\sin x})}=\lim_{x\to 0}\frac{\tan x-\sin x}{x^3(\sqrt{1+\tan x}+\sqrt{1+\sin x})}$. Step 2 · Factor Trigonometric Terms: $\tan x-\sin x=\frac{\sin x}{\cos x}-\sin x=\sin x\left(\frac{1-\cos x}{\cos x}\right)$. Multiply numerator and denominator by $1+\cos x$.

19 [Challenging] • Target: The chain rule in applied rate problems • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Level 3 · AdvancedLet $h(x)=\tanh(x)=\frac{e^x-e^{-x}}{e^x+e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

Step 1 · Algebraic Monotonicity: Rewrite $h(x)$ by dividing numerator and denominator by e^x:

$h(x)=\frac{1-e^{-2x}}{1+e^{-2x}}=\frac{(1+e^{-2x})-2e^{-2x}}{1+e^{-2x}}=1-\frac{2}{e^{2x}+1}$. As x, $\frac{2}{e^{2x}+1}\in(0,2)\Rightarrow h(x)=1-\frac{2}{e^{2x}+1}\in(-1,1)$. The domain of h^{-1} is $(-1,1)$. Set $y=\frac{e^{2x}-1}{e^{2x}+1}$: $y(e^{2x}+1)=e^{2x}-1\Rightarrow ye^{2x}+y=e^{2x}-1\Rightarrow e^{2x}(1-y)=1+y$. For $y\in(-1,1)$.

20 [Challenging] • Target: Chain rule with a table of composed values • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Level 3 · Advanced Find the extreme values of $f(x,y)=x^2+2y^2$ on the circle constraint $g(x,y)=x^2+y^2=1$ using the Method of Lagrange Multipliers.

Set up Lagrange system $ablaf=\lambda ablag$: $ablaf=\langle 2x,4y\rangle$, $ablag=\langle 2x,2y\rangle$. 1) $2x=2\lambda x\Rightarrow 2x(1-\lambda)=0$ 2) $4y=2\lambda y\Rightarrow 2y(2-\lambda)=0$ 3) $x^2+y^2=1$ From (1), either $x=0$ or $\lambda=1$: • If $x=0$: from constraint (3), $y^2=1\Rightarrow y=\pm 1$. Points: $(0,1)$ and $(0,-1)$. Value: $f(0,\pm 1)=0+2(1)^2=2$. • If $\lambda=1$: from (2), $2y(2-1)=2y=0\Rightarrow y=0$. From constraint (3).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic chain rule applications

Use the chain rule to differentiate $f(x)=(x-1)^2$, then find $f'(1)$.

a. 0 ✓

b. -2

c. 3

d. 5

$f'(x)=2(x-1)^1(1)$, so $f'(1)=2(0)^1(1)=0$.

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22 [Foundational] • Target: The chain rule with trig and exponential functions

Which is the correct derivative of $F(x) = \sin(5x + 2)$?

a. $5\cos(5x + 2)$ ✓

b. $-5\sin(5x + 2)$

c. $5\cos(x)$

d. $\cos(5x + 2)$

By the chain rule, $F'(x) = \cos(u) \cdot u'$ where $u = 5x + 2$ and $u' = 5$: $F'(x) = 5\cos(5x + 2)$.

23 [Foundational] • Target: Nested (multiple) chain rule

The side of a square is increasing at 2 cm/s. Find the rate of change of its area when the side length is 5 cm.

a. 18

b. 10

c. **20** ✓

d. 22

$A = s^2 \Rightarrow \frac{dA}{dt} = 2s\frac{ds}{dt} = 2(5)(2) = 20$ cm^2/s.

24 [Foundational] • Target: Chain rule combined with product/quotient rules

The radius of a circle increases at a rate of 3 cm/s. Find the rate of change of the area $\left(\frac{dA}{dt}\right)$ when $r = 17$ cm.

a. 34π

b. 102π ✓

c. 102

d. 51π

$A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi(17)(3) = 102\pi$ cm^2/s.

25 [Foundational] • Target: Chain rule with tables of values

The region between $y = 2$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{19}{5}$

b. $\frac{19}{10}\pi$

c. $\frac{19}{5}\pi$ ✓

d. $\frac{38}{5}\pi$

Washer method: $V = \pi\int_0^1 (2^2 - (x^2)^2)dx = \frac{19}{5}\pi$.

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26 [Foundational] • Target: Chain rule with inverse trig and logarithms

Find the x-coordinate of the inflection point of $f(x) = x^3 - 5x^2$.

a. $\frac{5}{3}$ ✓

b. $\frac{8}{3}$

c. $\frac{2}{3}$

d. $-\frac{5}{3}$

$f''(x) = 2(3x - 5)$. Setting $f''(x) = 0$ gives $x = \frac{5}{3}$, where concavity changes.

27 [Foundational] • Target: Finding tangent lines using the chain rule

The region under $y = 1\sqrt{x}$ on $[0, 4]$ is rotated about the x-axis. Find the volume of the resulting solid.

a. 16π

b. 8π ✓

c. 4π

d. 8

Disk method: $V = \pi\int_0^4 (1\sqrt{x})^2 dx = \pi\int_0^4 1x\,dx = 8\pi$.

28 [Foundational] • Target: Visualizing a composite function and its derivative

If $f(x) = 1x^2 + 3x$, is f increasing or decreasing at $x = 0$?

a. decreasing

b. cannot be determined

c. constant

d. **increasing** ✓

$f'(x) = 2x + 3$, so $f'(0) = 3$, which is positive. Thus f is increasing at $x = 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The chain rule in applied rate problems

Evaluate the definite integral of the constant function: $\int_8^{11} 21\,dx$.

a. 21

b. 84

c. 42

d. **63** ✓

The antiderivative is $21x$. Evaluating from 8 to 11 gives $21(11 - 8) = 63$.

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30 [Intermediate] • Target: Chain rule with a table of composed values

Find the area under the curve $f(x) = 1x^2 + 1$ on $[1, 2]$.

a. $\frac{16}{3}$

b. $-\frac{10}{3}$

c. $\frac{4}{3}$

d. $\frac{10}{3}$ ✓

Area $= \int_1^2 (1x^2 + 1)dx = \frac{10}{3}$.

31 [Intermediate] • Target: Basic chain rule applications

Find the critical number of $f(x) = 5x^2 + 0x$.

a. **0** ✓

b. -1

c. 1

d. 2

$f'(x) = 10x$. Setting $f'(x) = 0$ gives $x = 0$.

32 [Intermediate] • Target: The chain rule with trig and exponential functions

A function is continuous at $x = a$ if:

a. $f(a)$ exists

b. $\lim_{x \to a} f(x)$ exists

c. **Both exist and are equal** ✓

d. None of the above

Continuity requires both to exist and be equal.

33 [Intermediate] • Target: Nested (multiple) chain rule

Find the area bounded by $f(x) = 2x$, the x-axis, and the line $x = 6$, for $x \geq 0$.

a. 34

b. 38

c. 12

d. **36** ✓

Area $= \int_0^6 2x\,dx = 36$.

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34 [Intermediate] • Target: Chain rule combined with product/quotient rules

The diagram shows the graph of the function $y = f(x)$. The equation of the function is given by $f(x) = ax^2 + bx + c$. Given that the vertex of the parabola is at $(-1, 8)$ and the graph passes through the point $(1, 0)$, find the values of a and b.

a. **a = -2, b = -4 ✓**

b. a = -2, b = 4

c. a = 2, b = 4

d. a = 2, b = -4

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ is at $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$. We are given that the vertex is at $(-1, 8)$. Therefore, $-\frac{b}{2a} = -1$, which implies $b = 2a$. We are also given that the vertex is at $y = 8$, so $f(-1) = 8$. Substituting $x = -1$ into the function: $a(-1)^2 + b(-1) + c = 8 \Rightarrow a - b + c = 8$. Since $b = 2a$, we have $a - 2a + c = 8 \Rightarrow -a + c = 8$. The graph also passes through the point $(1, 0)$. So, $f(1) = 0$. Subst.

35 [Intermediate] • Target: Chain rule with tables of values

Consider the region R bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 0$ and $x = 1$. A solid is formed by rotating R about the y-axis. Determine the volume of this solid using the method of cylindrical shells. Express your answer in terms of π.

a. $\frac{\pi}{2}$

b. $\pi \ln(2)$

c. π ✓

d. $\frac{\pi}{4}$

The region R is defined by $y = \frac{1}{x^2+1}$, $y = 0$, $x = 0$, and $x = 1$. We are rotating this region about the y-axis. Using the method of cylindrical shells, the volume V is given by the integral: $V = \int_a^b 2\pi x h(x) dx$ where x is the radius of the cylindrical shell, and $h(x)$ is the height of the shell. In this case, the radius of a cylindrical shell at a distance x from the y-axis is simply x. The height of the shell is the value of the function $y = \frac{1}{x^2+1}$. The limits of integr.

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36 [Intermediate] • Target: Chain rule with inverse trig and logarithms

Consider a region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the y-axis. Let S be the solid generated when R is revolved about the y-axis. The volume of S can be calculated using the method of cylindrical shells or the disk/washer method. However, direct application of standard integration techniques to find the volume of S leads to an integral that cannot be evaluated in terms of elementary functions. A related problem involves finding the surface area of a solid of revolution. Let C be the curve defined by $y = \sqrt{1 - x^2}$ for $0 \le x \le 1$. This curve represents a quarter-circle. If this curve C is revolved about the x-axis, it generates a hemisphere. The surface area of this hemisphere (excluding the base) is given by the integral $A = \int_0^1 2\pi y \sqrt{1 + (y')^2}\, dx$. Determine the exact surface area of the generated hemisphere.

a. $\frac{\pi}{2}$

b. π ✓

c. 2π

d. 4π

The curve is given by $y = \sqrt{1 - x^2}$ for $0 \le x \le 1$. We first find the derivative of y with respect to x: $y' = \frac{d}{dx}(\sqrt{1 - x^2}) = \frac{1}{2\sqrt{1 - x^2}} \cdot (-2x) = \frac{-x}{\sqrt{1 - x^2}}$. Next, we compute $1 + (y')^2$: $1 + (y')^2 = 1 + \left(\frac{-x}{\sqrt{1 - x^2}}\right)^2 = 1 + \frac{x^2}{1 - x^2} = \frac{1 - x^2 + x^2}{1 - x^2} = \frac{1}{1 - x^2}$. Then, $\sqrt{1 + (y')^2} = \sqrt{\frac{1}{1 - x^2}} = \frac{1}{\sqrt{1 - x^2}}$ for $0 \le x < 1$. The surface area integral is.

37 [Intermediate] • Target: Finding tangent lines using the chain rule

At what rate is the area of a square changing when its side is 4 m and growing at 2 m/s?

a. 8 m²/s

b. **16 m²/s** ✓

c. 32 m²/s

d. 4 m²/s

Applying Applications of Derivatives principles, $dA/dt = 2s \cdot ds/dt = 2(4)(2) = 16$. Therefore, the verified correct option is 16 m²/s.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Visualizing a composite function and its derivative • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

The integral $\int_0^1 e^{-x^2}dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

b. $1 - 1/3 \approx 0.667$ ✓

c. $1 + 1/3 \approx 1.333$

d. 0

$\int_0^1 e^{-x^2}dx \approx 1 - 1/3 + \cdots \approx 0.667.$

39 [Challenging] • Target: The chain rule in applied rate problems • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Using the Product Rule, $\frac{d}{dx}[(3x+1)(x^2-2)] =$

a. $3(x^2-2) + 2x(3x+1)$

b. $3x^2 - 6 + 6x^2 + 2x$

c. $9x^2 + 2x - 6$

d. **all are equivalent** ✓

All three expressions are equivalent: $9x^2 + 2x - 6$.

40 [Challenging] • Target: Chain rule with a table of composed values • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

If $f(x) = \frac{4}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. **2** ✓

b. 1

c. 3

d. 4

$\frac{4}{x} = x \Rightarrow x^2 = 4 \Rightarrow x = 2$

41 [Challenging] • Target: Basic chain rule applications • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

A function has exactly one local min and no local max on (a, b). Its f' must change from:

a. $-$ **to** $+$ ✓

b. $+$ to $-$

c. $+$ to $+$

d. 0 to $+$

Local min: f' negative then positive.

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42 [Challenging] • Target: The chain rule with trig and exponential functions • Cross-Lesson: 7.01 Differential Equations and Slope Fields

The graph of the function $f(x) = ax^2 - 6x + c$ is shown below. The graph has a vertex at the point (3, -9) and passes through the point (0, 0). Determine the values of a and c.

a. $a = 1, c = 0$

b. $a = 2, c = 0$ ✓

c. $a = 1, c = 3$

d. $a = 2, c = 3$

The function is given by $f(x) = ax^2 - 6x + c$. The vertex of a parabola in the form $f(x) = Ax^2 + Bx + C$ is located at $x = -\frac{B}{2A}$. In this case, $A = a$ and $B = -6$. Therefore, the x-coordinate of the vertex is $x = -\frac{-6}{2a} = \frac{6}{2a} = \frac{3}{a}$. We are given that the vertex is at (3, -9), so the x-coordinate of the vertex is 3. Equating the two expressions for the x-coordinate of the vertex: $\frac{3}{a} = 3$ Multiplying both sides by a gives $3 = 3a$, so $a = 1$. Now, we use the y-c.

43 [Challenging] • Target: Nested (multiple) chain rule • Cross-Lesson: 8.01 Area Between Curves

The graph of the function $f(x) = \frac{x^2 - ax + 6}{x - 1}$ is shown below. The graph has a vertical asymptote at $x = 1$ and a point of discontinuity at $(2, k)$. Find the value of a.

a. **4** ✓

b. 5

c. 6

d. 7

The function is given by $f(x) = \frac{x^2 - ax + 6}{x - 1}$. $x = 1$ is a vertical asymptote, which means that $x - 1$ is not a factor of the numerator, and when $x = 1$, the numerator is non-zero. The point of discontinuity is at $(2, k)$. This implies that $x = 2$ is a root of the numerator, so $f(2) = 0$ or the expression is undefined at $x = 2$ if $x - 2$ were a factor of the denominator. Since the denominator is $x - 1$, which is not zero at $x = 2$, the function must be zero at $x = 2$ for there to be a removable discontinuity (a.

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44 [Challenging] • Target: Chain rule combined with product/quotient rules • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

The graph of the function $y = f(x)$ is shown below. The function is of the form $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(2, -4)$. The parabola passes through the points $(0, 0)$ and $(4, 0)$. Determine the value of a.

a. **-1** ✓

b. 1

c. -2

d. 2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. In this case, the vertex is $(2, -4)$, so $h = 2$ and $k = -4$. Thus, the function can be written as $f(x) = a(x - 2)^2 - 4$. We are given that the parabola passes through the point $(0, 0)$. Substitute these coordinates into the vertex form equation: $0 = a(0 - 2)^2 - 4$ $0 = a(-2)^2 - 4$ $0 = 4a - 4$ $4a = 4$ $a = 1$. Let's verify with the other given point.

45 [Challenging] • Target: Chain rule with tables of values • Cross-Lesson: 1.01 Limits Graphically and Numerically

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. What are the coordinates of point A?

a. **(2, 0)** ✓

b. (4, 0)

c. (0, 8)

d. (1, 3)

To find the x-intercepts of the function $f(x) = x^2 - 6x + 8$, we need to solve for x when $f(x) = 0$. This means we need to find the roots of the quadratic equation $x^2 - 6x + 8 = 0$. We can factor this quadratic equation. We are looking for two numbers that multiply to 8 and add to -6. These numbers are -2 and -4. Therefore, the equation can be factored as $(x - 2)(x - 4) = 0$. Setting each factor to zero gives us the x-intercepts: $x - 2 = 0 \Rightarrow x = 2$ and $x - 4 = 0 \Rightarrow x = 4$. The graph sho.

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